PVP 14

Code: CE2T3, CS2T3, IT2T3

I B.Tech - II Semester – Regular Examinations – JULY 2015

ENGINEERING PHYSICS
(Common for CE, CSE & IT)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks
11x 2 = 22 M

1. a) What are the properties of matter waves?

   b) Write an expression for plank's law of black body radiation and explain the terms?

   c) What are Eigen values?

   d) Define i) Unit cell ii) Space lattice

   e) Write notes on Bravais lattice and non-Bravais lattice?

   f) What do you mean by population inversion?

   g) What is Fermi Dirac distribution functions?

   h) What are different polarizations in dielectric materials?

   i) Define

      i) Magnetic Flux density ii) Magnetic Susceptibility

   j) What is indirect band gap in semiconductors and explain?

   k) Define Fermi energy level.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks. 3 x 16 = 48 M

2. a) Describe G. P. Thomson experiment setup and results, which gave the proof for existence Matter waves? 12 M

   b) In Davisson and Germer experiment, the second order diffraction set up in a constructive interference was observed at an angle of 40° with respect to electron beam direction for the wave length of electron waves 2.5A°. Calculate the inter planer distance of the experimental crystal. 4 M

3. a) State and derive the Bragg's law. 4 M

   b) Explain how Laue method can be use to determine the inner planar spaces. 8 M

   c) The minimum order of Bragg's reflection occurred at an angle of 20° in the plane [212]. Find the wavelength of x-ray if lattice constant is 3.615A°. ($d_{(212)}$=1.205 A°) 4 M

4. a) Derive the expression local uniform field of a dielectric material. Hence, obtain the expression of Clausius-Mossotti's equation. 9 M

b) Explain different types of polarization mechanisms. 7 M

5. a) Explain about the intrinsic Semiconductors. 4 M

b) Derive the expression for fermi energy level with concentration of holes and electrons in the conductive and valence band for intrinsic semiconductor. 10 M

c) For an intrinsic semiconductor gap width Eg=0.7ev calculate the Fermi level intrinsic semiconductor at 300 K ($m_r^* = m_e^* = m_0$) 2 M

6. a) Write the principle of semi conductor laser. 4 M

b) Describe the construction and working of Semi conductor laser. 8 M

c) Write a note on application of lasers. 4 M

PVP 14

Code: CE2T3, CS2T3, IT2T3

I B.Tech - II Semester – Regular Examinations – April 2016

ENGINEERING PHYSICS
(Common for CE, CSE & IT)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11x 2 = 22 M

1. a) What are the conclusions of de- Broglies Hypothesis?
   b) What is Heisenberg uncertainity principal?
   c) Explain the significance of Miller indices.
   d) Define and explain the space lattice.
   e) Mention the important failures of Classical free electron Theory.
   f) Define Dielectric constant.
   g) Explain the concept of Feerimagnetism.
   h) Write about Fermi level in intrinsic the Semi conductor.
   i) Explain the Population inversion condition.
   j) Write the principle of an Optical Fibre.
   k) What are nanomaterials.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2. a) Describe the construction and working of Davison and Germer experiment to estimate the dual nature of Material particle. 12 M

b) Calculate the de- Broglies wave length of an electron which has been accelerated from rest on application of potential of 400 volts. 4 M

3. a) Explain a short note on seven types of Crystal systems. 6 M

b) Find the packing fraction of SC, BCC, and FCC crystals. 6 M

c) A beam of X-rays is incident on a Nacl crystal with lattice spacing 0.282 nm. Calculate the wavelength of X- rays if the first order Bragg reflection takes place at a glancing angle of $8^{o}\ 35'$. 4 M

4. a) Derive the expression for the electronic polarisability of a dielectric material 10 M

b) Derive the Classius-Mossotti relation. 4 M

c) The dielectric constant of He gas at NTP is 1.0000684. Calculate the electronic polarisability of He atoms if the gas contains 2.7×10^{25} atoms/m^3. 2 M

5. a) Distinguish between intrinsic and extrinsic semiconductor. 6 M

b) Derive a expression for the carrier concentration of an n-type semiconductor. 6 M

c) Discuss Fermi level in p type semiconductors. 4 M

6. a) Discuss the electrical and mechanical properties of nanomaterials. 6 M

b) Explain the synthesis of nanomaterials by Sol-Gel method with a neat flow chart. 6 M

c) Write any two applications of LASERS in defense. 4 M

PVP 14

Code: CE2T3, CS2T3, IT2T3

I B.Tech - II Semester – Regular/Supplementary Examinations - May 2017

ENGINEERING PHYSICS
(Common for CE, CSE & IT)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11x 2 = 22 M

1.

a) The kinetic energy of an electron of wavelength 3×10^{-8}m in terms of eV is?

b) A particle is trapped in one dimensional box of width $2X10^{-9}$m along x-axes. What is the Eigen value of the particle, if the particle is present in its 2nd energy level.

c) The miller indices of a plane that makes intercepts of a, 2b, and 5c on the crystallographic axes of an orthorhombic crystal.

d) The Bragg's angle of diffraction for its second order (110) reflection of cubic crystal of rock salt with lattice parameter 2.81A˚, if X-rays of wavelength 0.71A˚ are used?

e) What is the lattice parameters configuration of a Tetragonal and Hexagonal crystal system?

f) Define polarisation and polarisability.

g) Define Internal Field in Dielectric materials.
h) What are distinguishing features of ferromagnetism?
i) What is spontaneous emission in LASERS?
j) List any two attenuations in Optical fibers.
k) Describe Nano tubes.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2. a) What is Heisenberg's uncertainty principle? 3 M

b) Explain the matter waves. Describe how G.P Thomson experiment was verified the existence of matter waves. 8 M

c) Explain the physical significance of wave function. 5 M

3. a) What is Bravais lattice? Give the lattice parameters configuration of fourteen Bravais lattices among the seven crystal systems and draw any three of them neatly. 8 M

b) What are miller indices and their significances? 4 M

c) Show that in a simple cubic lattice; inter planar spacing between the successive lattice planes (100), (110) and (111) are in the ration of 1:0.71:0.58. 4 M

4. a) Explain the Bloch Theorem. 5 M

b) Give the qualitative treatment of Kroning-Penny model for energy bands. Based on the band theory of solids, distinguish between conductors, semiconductors and insulators. 11 M

5. a) Write a short note on soft and hard magnetic materials. 6 M

b) Derive the expression for the drift and diffusion currents. 6 M

c) Derive the Einstein relations. 4 M

6. a) Illustrate construction and working of He-Ne Gas laser. Write any two applications of laser? 10 M

b) Derive the expression for acceptance angle and Numerical Aperture in Optical Fibers. 6 M

PVP 14

Code: CE2T3, CS2T3, IT2T3

I B.Tech - II Semester – Regular/Supplementary Examinations
April - 2018

ENGINEERING PHYSICS
(Common for CE, CSE & IT)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11x 2 = 22 M

1.

a) Give an account of Heisenberg's Uncertainty principle.
b) Can matter waves travel faster than light? Justify your answer.
c) Why X-rays are used for crystal structure analysis?
d) What are lattice parameters?
e) Draw the E-K diagram for free electrons at different values of k, indicating the allowed and forbidden bands.
f) What is atomic polarisability? What is the relation between dipole moment and atomic polarisability?
g) What are direct and indirect band gap materials?
h) Define magnetic susceptibility (χ) and relative permeability (μ_r). Write the relation between them.
i) Differentiate between spontaneous and stimulated emission processes.

j) Explain total internal reflection. What are the conditions to have T.I.R?

k) What is quantum confinement in nano materials?

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2. a) Obtain the time independent Schrodinger wave equation for a free particle in a one dimensional box. 8 M

b) Derive the expression for de-Broglie wavelength of an electron accelerated through a potential difference. 4 M

c) Calculate the de-Broglie wavelength of electron which has kinetic energy of 15 eV. 4 M

3. a) Evaluate the atomic radius of S.C, B.C.C, and F.C.C. structures. 6 M

b) Discuss briefly the Laue experimental method for crystal structure determination by X-ray diffraction. 5 M

c) X-rays of wavelength 1.5 A^o makes a glancing angle of 60° in the first order when diffracted from (111) plane of NaCl crystal. Find the lattice constant of NaCl. 5 M

4. a) What is the effect of periodic potential on the energy of electrons in a metal? Explain it on the basis of Kronig Penney model and explain the formation of energy bands. 6 M

b) Discuss different types of polarizations in dielectrics. 6 M

c) If an ionic crystal is subjected to an electric field of 1000 Vm^{-1} and the resulting polarization 4.3×10^{-8} cm^2. Calculate the relative permittivity of NaCl. 4 M

5. a) Derive the expression for the carrier density in an n-type semiconductor and the position of Fermilevel in n-type Semiconductor. 6 M

b) Discuss about diamagnetic, paramagnetic and ferromagnetic substances citing one example of each. 6 M

c) The magnetic susceptibility of aluminum is $2.3x10^{-5}$. Find its permeability and relative permeability. 4 M

6. a) Discuss with suitable diagram the principle, construction and working of He-Ne LASER. Describe any three applications of LASER. 6 M

b) Explain the terms Numerical aperture and acceptance angle of a fiber. Derive the expression for them. 6 M

c) Calculate the Numerical aperture and acceptance angle of optical fiber of refractive indices for core and cladding 1.62 and 1.52 respectively. 4 M